

STONE ČECH COMPACTIFICATION

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1. MOTIVATION

Given a top space X , we want the *best possible* approximation of X that is compact and hausdorff. Consider the left adjoint of the inclusion functor from category of compact hausdorff spaces with continuous functions to the category of topological spaces with continuous functions. Let this left adoint functor be F . By the definition of left adointness we get

$$Hom_{\text{CompactHausdorff}}(F(X), K) = Hom_{\text{Top}}(X, K)$$

. This gives a universal property that every continuous function on X can be uniquely extended to $F(X)$.

2. PARTIAL ORDER

The obvious way to construct a compactification¹ is by embedding in a compact space² and taking the closure of the image, but this need not necessitate that all continuous functions extend, for example $(0, 1)$ and $f(x) = \sin(1/x)$ is a continuous function from $(0, 1)$ to $\mathbb{R}^2$ that cannot be extended to $[0, 1]$. The set of compactifications is therefore partially ordered by the sets of continuous functions they extend. This can be equivalently stated as $K_1 \leq K_2$ iff there exists a surjective continuous map $q : K_2 \rightarrow K_1$.

3. CONSTRUCTION

Once we have the poset, the existence of the stone čech compactification depends on the existence of the *maximum* element. In language of order theory, the set of all compactifications of a tychonoff space X forms a complete upper semilattice. Let the set of all compactifications be $\mathcal{K} = \{(K_i, e_i)\}_{i \in I}$. Let $P = \prod_{i \in I} K_i$ ³ and $E(x) = (e_i(x))_{i \in I}$. E is an embedding and we define $K_{max} = \overline{E(X)}$. K_{max} dominates K_i for all $i \in I$ as we have $\Pi_i : P \rightarrow K_i$ and $\Pi_i|_{K_{max}} : K_{max} \rightarrow K_i$ and it is an easy exercise to show this map is surjective and fixes X .

4. LOOKING AT IT VIA LIMIT POINTS

If $K_1 \leq K_2$ then $q : K_2 \rightarrow K_1$, we can show $q(K_2 \setminus X) = K_1 \setminus X$.⁴ The obvious thing to try now is: to collapse a subset of $K \setminus X$ to a point and use the quotient topology on the new space to get a new compactification which is $\leq K$. The issue arises at hausdorffness, which in general does not behave well with quotient maps. We can collapse a subset of the boundary to a point iff it is closed in the overarching space⁵. An additional result is that a space is open in its compactification iff it is locally compact. Therefore a one point compactification exists iff the space is locally compact.

¹whenever compactification is used it means the new space is compact and hausdorff.

²An embedding in a compact space exists only for tychonoff spaces. This forces the space X to be tychonoff.

³for the sake of thouroughty, we are using tychonoff's theorem here to say the product is compact.

⁴The converse is obviously false as you would need the extended function between compactifications to be continuous.

⁵We are using the following general result here: The quotient space X/A (formed by identifying all points in A to a single point $[A]$) is a Hausdorff space if and only if A is closed in X , and for every $x \notin A$, there exist disjoint open sets $U, V \subseteq X$ such that $x \in U$ and $A \subseteq V$.

5. ON TYCHONOFF'S CUBE

Moving up the partial order keeps boundary points separate. Any two boundary points can be separated via a continuous function to $[0, 1]$. We embed X into the product space:

$$P = \prod_{f \in J} [0, 1]$$

where J is the set of all continuous functions from X to $[0, 1]$ via the evaluation map $e(x) = (f(x))_{f \in J}$. Let K be a compactification of X . Because K is a compact hausdorff space it can be embedded into a tychonoff cube indexed by $J_K = C(K, [0, 1])$. So, K sits faithfully inside $\prod_{g \in J_K} [0, 1]$. Every continuous function g on K restricts to a continuous function f on X . Therefore, J_K is just a strict subset of J (all bounded functions on X), hence there is a natural, continuous projection map:

$$\pi : \prod_{f \in J} [0, 1] \rightarrow \prod_{g \in J_K} [0, 1]$$

When you restrict this projection π to the closure space K_{max} , it maps perfectly onto K .

$$q = \pi|_{K_{max}} : K_{max} \rightarrow K$$

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6. APPENDIX

Theorem 1: A space is locally compact iff it is open in its neighbourhood